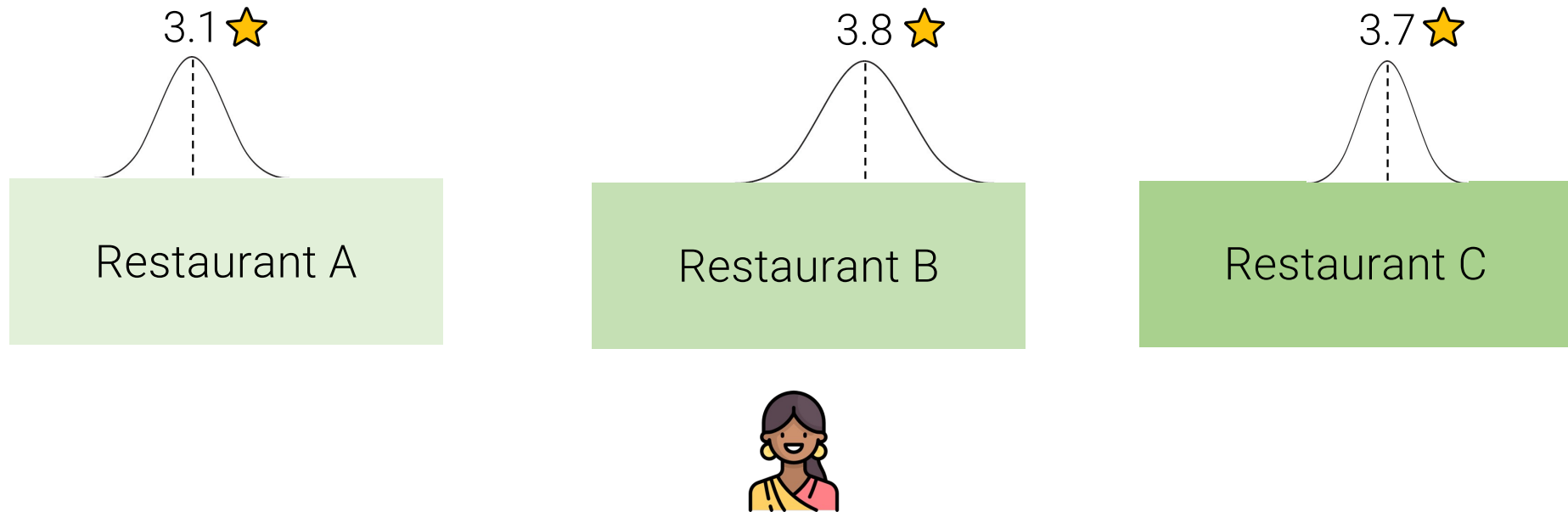
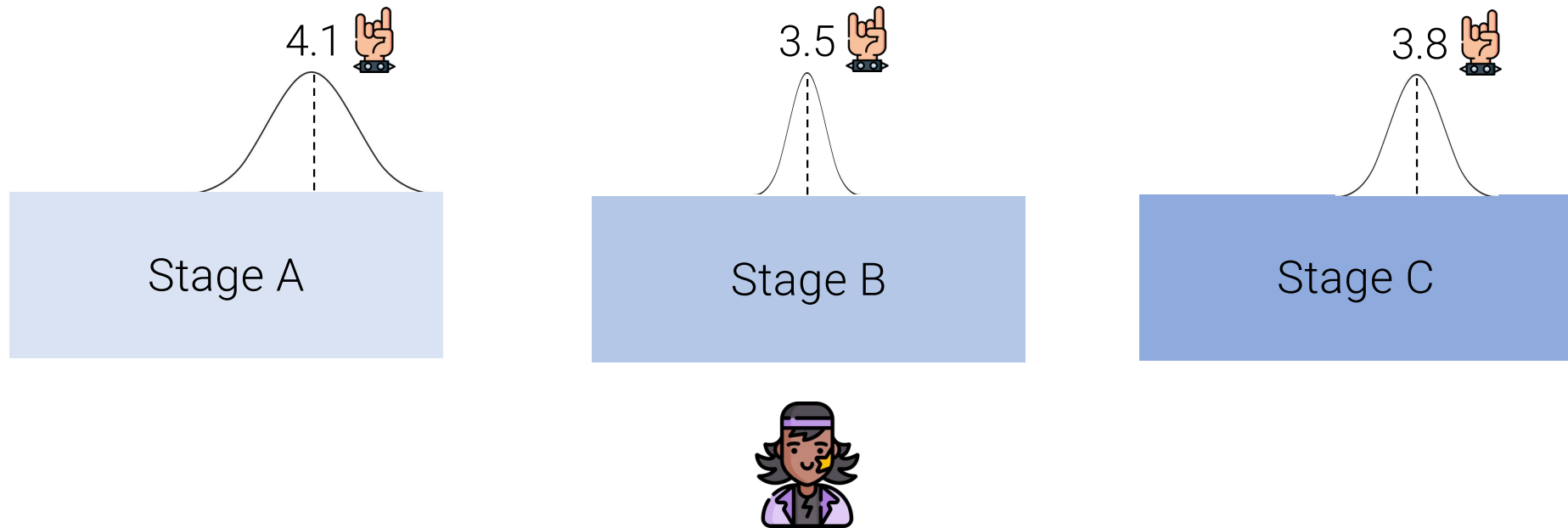


Exploration-exploitation scenarios



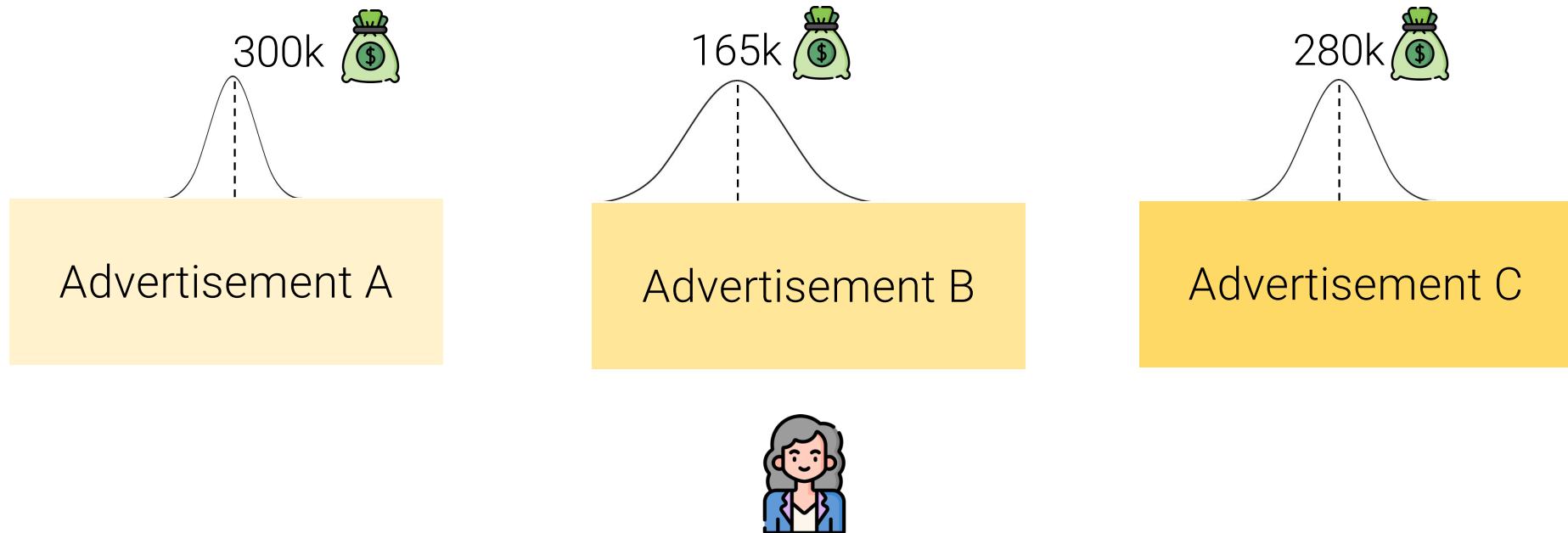
- Businesswoman visiting a town for 60 days
- The distribution of quality of dishes in the town restaurants is unknown
- **Objective:** dine every night in one of the many available restaurants such that her total satisfaction is maximized

Exploration-exploitation scenarios



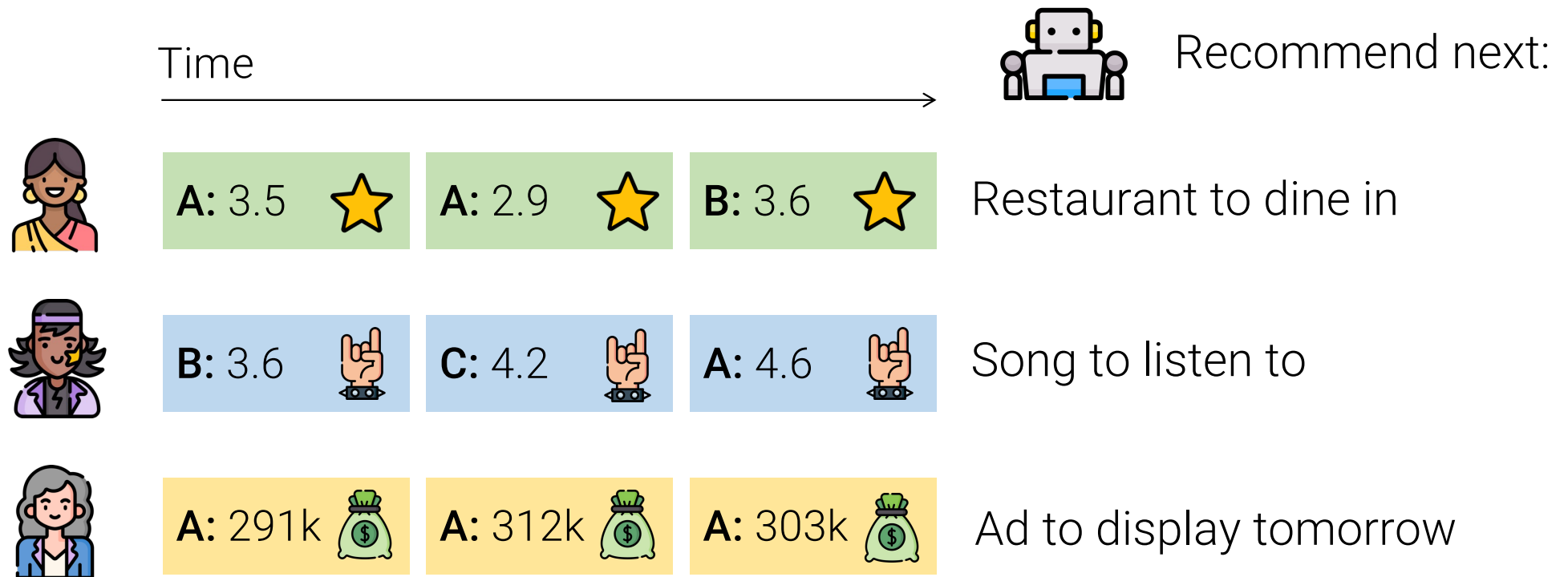
- Music enthusiast going to a music festival
- 50 songs are played throughout the day on each of the three stages in parallel
- The bands are all new to him, no idea how much he'll like them
- **Objective:** maximize the total satisfaction at the end of the day

Exploration-exploitation scenarios



- Advertiser prepared three different ads for her company
- Bought ad space on news website for 100 days, can show 1 ad per day
- The expected number of clicks per ad is unknown in advance
- **Objective:** maximize the total number of clicks on ad (revenue)

The recommender system perspective



Such that objective function is maximized

Bandits – definitions

- $B = \{(a_1, R_1), \dots, (a_K, R_K)\}$. Arm a_i and its distribution of rewards R_i
- $a_i(t)$. Reward from arm a_i at round t , ($0 < t \leq T$)
- $\mu(a_i) = \mathbb{E}[R_i]$. Expected value (mean) of the distribution of rewards R_i
- $\hat{\mu}_t(a_i) = \frac{1}{t} \sum_{i=1}^t a_i(t)$. Average empirical reward of arm a_i up to round t
- $\mu^* = \max_{i \in [1, K]} \mu(a_i)$. Best mean reward (multiple arms can have it)
- $\Delta(a_i) = \mu^* - \mu(a_i)$. Gap of arm a_i (how much it is worse than best)
- $a^* \mid \mu(a_i) = \mu^*$. Optimal arm

Strategy (policy) and regret

- A strategy (or policy) is an algorithm that defines the criterion for arm selection, possibly based on the observation of past rewards
- *“Pick a random arm and pull it for all rounds”* is a strategy

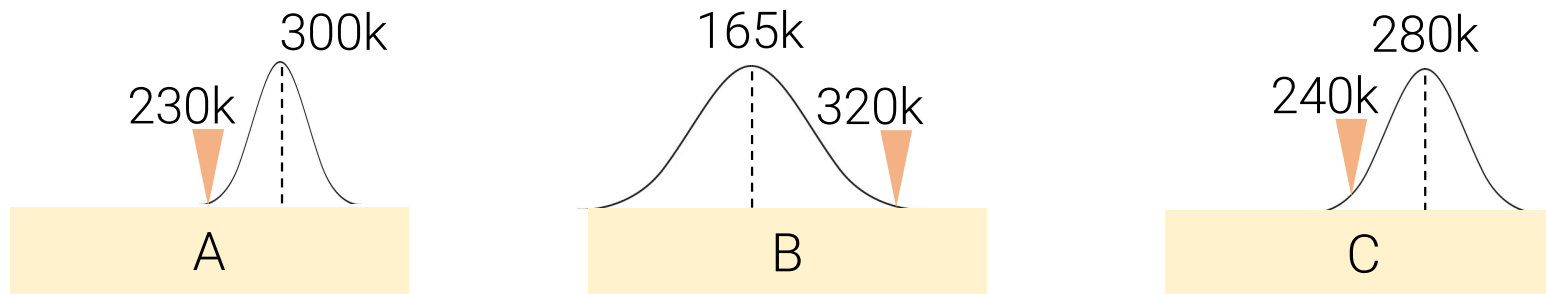
$$\rho(T) = (\mu^* \cdot T) - \sum_{t=1}^T a_i(t) \quad \text{Regret, namely loss wrt optimal-arm strategy}$$

$\mathbb{E}[\rho(T)]$ Expected regret for a certain strategy

$\rho(T) = 0, \text{ with } T \rightarrow \infty$ Zero-regret strategy

Basic strategies

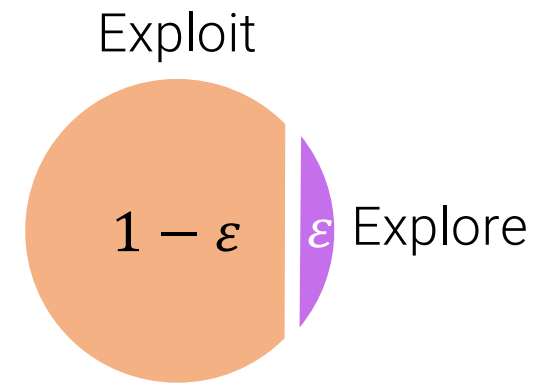
- Explore only (random)
 - Pick a random arm every time
- Exploit only (greedy)
 - Try all arms, then stick to the one with highest reward



An unlucky draw can get you stuck with the worst arm forever

ϵ -greedy strategy

- Start by pulling a random arm
- At each round t :
 - Pull the best arm so far ($\max \hat{\mu}_t(a_i)$) with probability $(1 - \epsilon)$
 - Pull a random arm with probability ϵ
- Variants
 - ϵ -first: to allow for a good degree of initial exploration, pull random arms for the first m rounds, then switch to ϵ -greedy.
 - ϵ -decay: the value of ϵ decays over time, as the need for exploration decreases once the probability of having found a good solution is high



Upper Confidence Bound (UCB1)

- At each round t , pull the arm that maximizes:

Natural log of current round

Number of samples of arm a_i up to round t

$$\underbrace{\hat{\mu}_t(a_i)}_{\text{Exploitation}} + \underbrace{\sqrt{\frac{2 \cdot \ln(t)}{N_t(a_i)}}}_{\text{Exploration}}$$

Average reward so far (equivalent to the exploit case of ϵ -greedy)

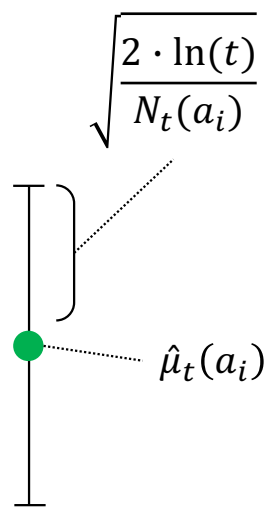


Diagram illustrating the UCB1 formula components:

- Exploitation:** $\hat{\mu}_t(a_i)$ (Average reward so far, equivalent to the exploit case of ϵ -greedy)
- Exploration:** $\sqrt{\frac{2 \cdot \ln(t)}{N_t(a_i)}}$ (Upper confidence bound term)

- Exploration term:
 - Grows for less-explored arms (lower denominator)
 - Decreases with each new sample, meaning that arms with high exploitation score will lose advantage over time, compared to unexplored arms

Thompson Sampling

$$p(\mu(a)|a(1), \dots, a(t)) \sim \mathcal{N}(\mu_P, \sigma_P)$$

$$\sigma_P^2 = \left(\frac{1}{\sigma_{max}} + t \right)^{-1}$$

$$\mu_P = \sigma_P^2 \cdot \sum_{\tau=1}^t a(\tau)$$



$a_1: ---$

$a_2: ---$

$a_1: 3.5$



$a_2: ---$

$a_1: 3.5$



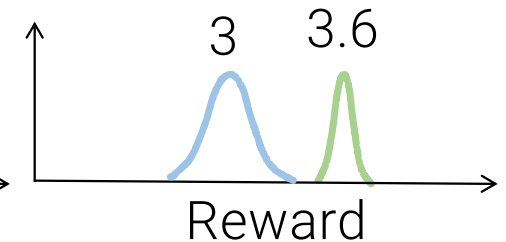
$a_2: 3.0$



$a_1: 3.5, 3.7$

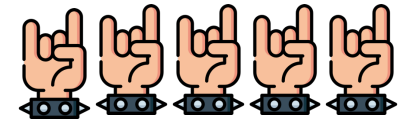


$a_2: 3.0$



Extending bandits with features

- Bandits explore new arms blindly
- No user/item features are considered
- Rewards could be used to learn features that are associated with high rewards
- Example:
 - User gets “The Number of the Beast” by Iron Maiden as recommended album
 - User likes it very much (high reward)
 - The system could learn that recommending “Powerslave”, an album from the same band, may lead to a similar reward



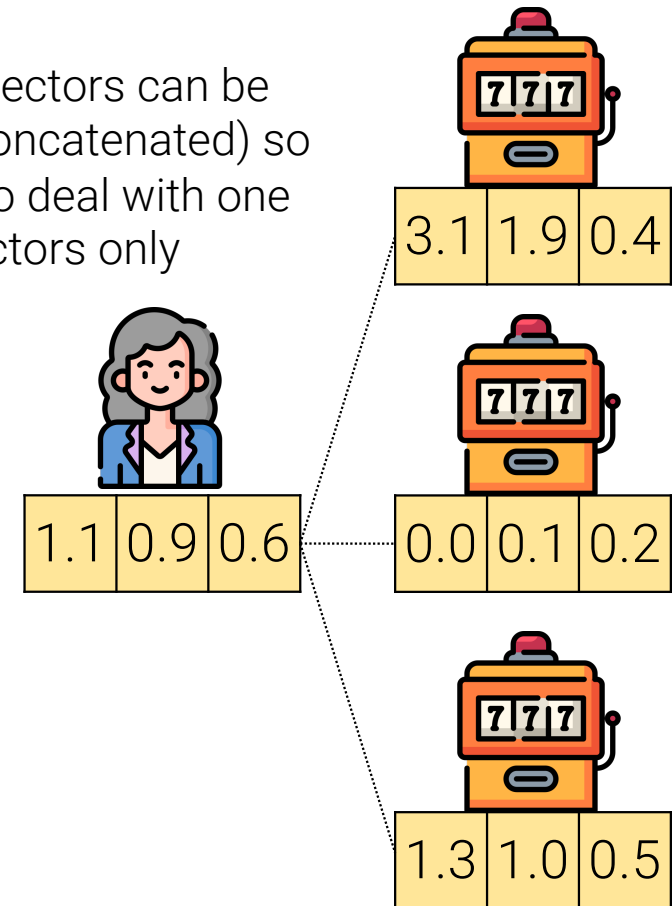
???



Contextual bandits

- Context = feature vector
- Shared, when it is common to all arms
 - User profile (age, preferences from past interactions)
 - Global context (time of day, location, etc.)
- Action-specific, when it is associated to a specific arm
 - Content features of a song
 - Position of an ad in the page

User and item vectors can be combined (e.g., concatenated) so that one needs to deal with one context vectors only



Linear UCB

- Assumption: the expected payoff of an arm is a linear combination of its feature vector

$$\mathbb{E}[a(t) | \vec{x}_{t,a}] = \vec{x}_{t,a}^T \cdot \theta_a$$

- An arm a_t at step t is selected to maximize:

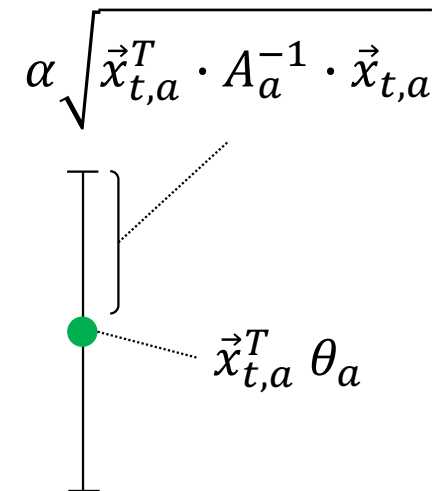
$$a_t = \operatorname{argmax}_{a \in \mathcal{A}} \left(\underbrace{\vec{x}_{t,a}^T \cdot \theta_a}_{\text{Exploitation}} + \underbrace{\alpha \sqrt{\vec{x}_{t,a}^T \cdot A_a^{-1} \cdot \vec{x}_{t,a}}}_{\text{Exploration}} \right)$$

Exploitation

Exploration parameter
(higher = more exploration)

Exploration

Bayesian inference derivation



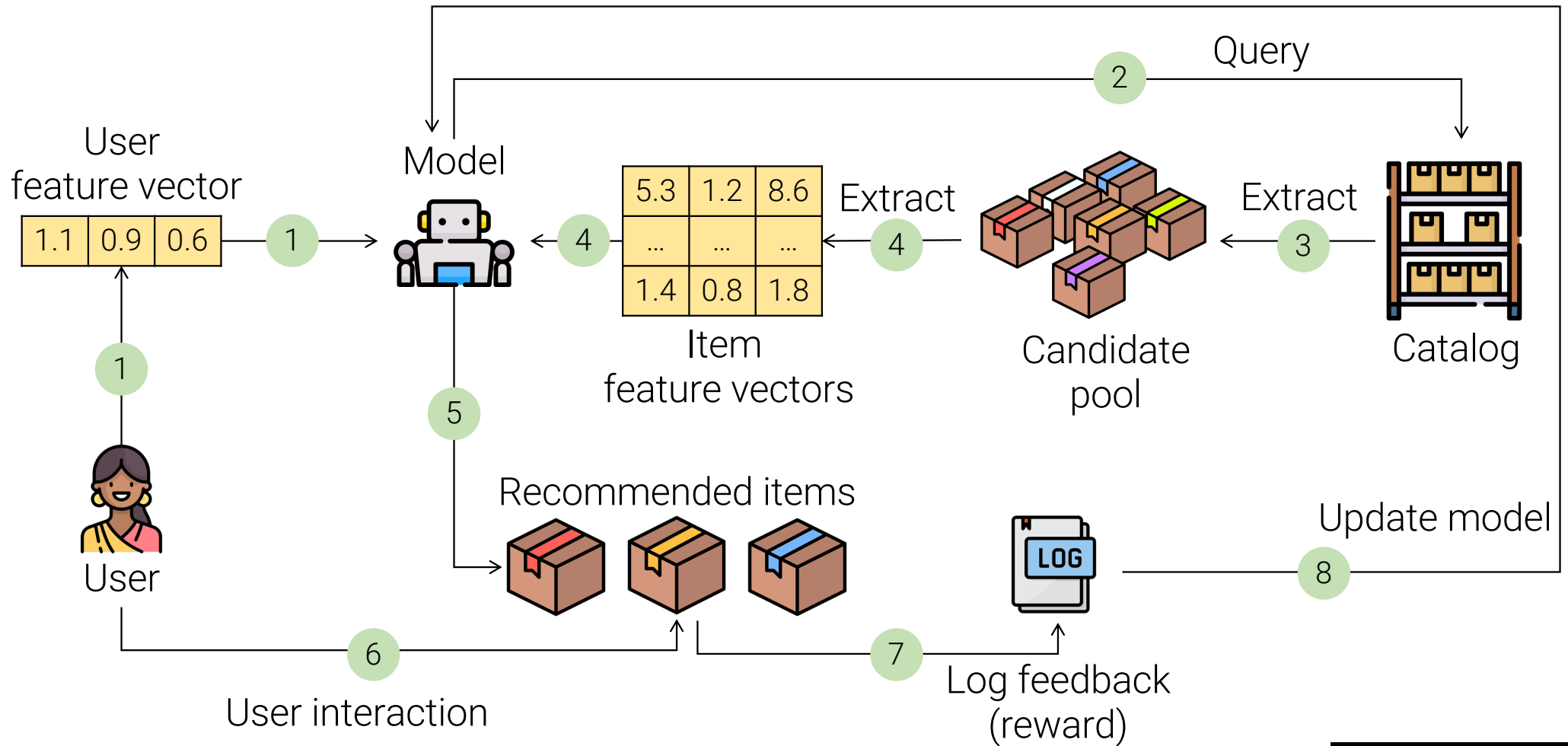
Linear UCB algorithm (pseudocode)

```
for  $t$  in range( $1 \dots T$ ):  
  for all  $a \in \mathcal{A}$ :  
    if  $a$  is new:  
       $A_a = \mathbb{I}_d$  #  $d$ -dimensional identity matrix  
       $b_a = \mathbf{0}_{d \times 1}$  #  $d$ -dimensional zero vector  
       $\theta_a = A_a^{-1} \cdot b_a$   
  
      
$$p_{t,a} = \vec{x}_{t,a}^T \cdot \theta_a + \alpha \sqrt{\vec{x}_{t,a}^T \cdot A_a^{-1} \cdot \vec{x}_{t,a}}$$
  
    choose arm  $a_i$  with  $\max p_{t,a}$   
    observe payoff  $a_i(t)$   
     $A_a = A_a + \vec{x}_{t,a} \cdot \vec{x}_{t,a}^T$   
     $b_a = b_a + a_i(t) \cdot \vec{x}_{t,a}$ 
```

Linear version of
Thompson Sampling
(LinTS) exists

Li et al "A Contextual-Bandit Approach to Personalized News Article Recommendation", 2010
<https://arxiv.org/abs/1003.0146>
Agrawal, Goyal "Thompson Sampling for Contextual Bandits with Linear Payoffs", 2013
<https://proceedings.mlr.press/v28/agrawal13.pdf>

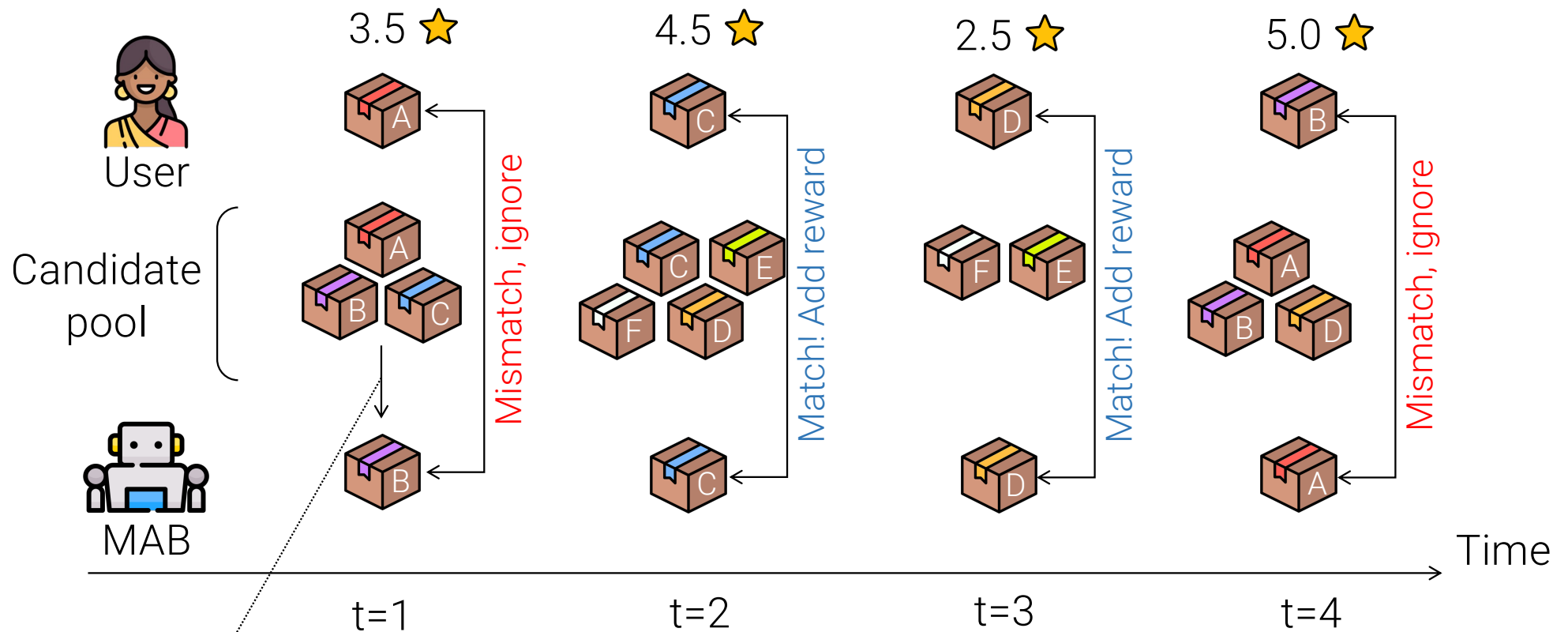
Bandit recommendation workflow



Replay evaluation

History = C, D

Total reward = $4.5 + 2.5 = 7$



MAB selects arm based on policy and previous user history (up to t-1)

Replay recap

- Run the system to obtain a log with tuples:
 - $(time, user, arm, reward, context, candidate\ pool)$
 - arm is the one shown to the user, selected from the $candidate\ pool$ with some strategy (ideally, at random)
- Start with an empty history and list of rewards
- For each time step t , ask MAB to pull an arm based on history
- If the arm pulled by MAB at step t matches with the one selected by the user at time t , then add the arm to the history and the reward to the cumulative reward. Else, go to $t+1$ without performing any other action

Multiarmed bandits libraries

- <https://www.kaggle.com/getting-started/131811>
- <https://pypi.org/project/SMPyBandits/>
- <https://github.com/HCDM/BanditLib>
- <https://github.com/bgalbraith/bandits>
- <https://github.com/roycoding/slots>
- <https://github.com/fidelity/mabwiser>
- <https://github.com/johnmyleswhite/BanditsBook>

Exercise

- Implement the UCB strategy and test it with Replay on a synthetic dataset

